

Practical multistate models

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Introduction

Multistate time to event models based on hazards, i.e., multistate hazard models (MSH) have been an extremely useful tool for the medical research teams that I have been involved in at Mayo Clinic. See for instance Jack et al, PMID:35310829 and PMID: 41240917, Allen et al PMID: 41240917 and PMID: 28941364 and Le-Rademacher PMID: 36003089 as examples. They are able to capture multiple aspects of a disease process, and thus can be an important tool for deeper understanding.

This document is a “nuts and bolts” description that reprises some of those analyses, with a focus on the data preparation, data checking, and analysis code needed. Some mathematics will appear, but only to help clarify the models.

For all the models, key first steps are * What data do we have, and what questions *could* be answered with that data? * The overall goal: what aspects of the process(es) do we wish to describe, and in relation to what predictors? No paper can do everything. * Lay out the multi-state model as a diagram. Refer to it often. * Create the analysis data set, with particular attention to questions of immortal time bias. * Look at the data (look, not just glance) and decide what to do with special cases. There are invariably one or two.

Mathematically we describe our multistate model in terms of three processes. * $N_{ijk}(t)$ = the number of observed transitions from state j to state k , up to time t for subject i . * $Y_{ij}(t)$ equals 1 if subject i is currently in state j , at time t , and at risk for transition to another state at that time. Subjects who have entered an absorbing state prior to t , e.g., death, are for instance not at risk for other transitions. * $X_i(t)$ = the set of covariates for subject i at time t . Most often, most of these covariates will remain static, e.g., treatment arm or sex, while others are time-dependent, e.g. cholesterol might be remeasured at each visit.

There is a subtle difference between “at risk for an event” and “at risk for an observed event”, i.e., one that will be, if it occurs, captured in the data. If subject i has incomplete follow-up, “censoring”, then $Y_i(t)$ for observed events will become 0 sometime before they enter an absorbing state. In this document I will largely ignore this, and use Y, N, X to refer to observed data. If censoring is not random, however, then our estimators can be biased, and any investigation into the hows and whys of that bias will need to carefully distinguish them.

Immortal Time Bias

In our opinion, one of the greatest threats to valid medical research is the presence of immortal time bias (ITB). In short, for a valid time-to-event model none of $N(t)$, $Y(t)$ nor $X(t)$ can contain values that depend on the future. In our personal experience, MSH models provide, unfortunately, even more opportunities for this error, ones which are sometimes difficult to spot. The number of ways to create ITB appears to be near infinite. Here are a few examples.

fill in later

Timeline data

The survival package supports two forms of multistate data. The older and better known one is referred to as counting process form. The first 3 lines below show the myeloid data in its raw form as found in the survival package. The next lines show the timeline form. The code to create mdata1 will come later.

```
subset(myeloid, id<4)
```

```
##   id trt sex flt3 futime death txtime crtime rltime
## 1  1  B  f   C   235     1    NA     44    113
## 2  2  A  m   B   286     1   200     NA     NA
## 3  3  A  f   A  1983     0    NA     38     NA
```

```
subset(mdata1, id<4)
```

```
##   id trt sex flt3 tstart tstop death cr sct relapse
## 1  1  B  f   C     0    44     0  1  0         0
## 2  1  B  f   C    44   113     0  0  0         1
## 3  1  B  f   C   113   235     1  0  0         0
## 4  2  A  m   B     0   200     0  0  1         0
## 5  2  A  m   B    200   286     1  0  0         0
## 6  3  A  f   A     0    38     0  1  0         0
## 7  3  A  f   A    38  1983     0  0  0         0
```

Each row of data in the timeline form consists of a subject id, a time interval (tstart, tstop], the event(s) if any that occur (death, cr, sct, relapse), and covariates (trt, sex, flt3). Covariate apply over the entire interval (tstart, tstop], and events happen at the end of the interval. In this particular data set there are no time-dependent covariates, and multiple possible endpoints: complete response, stem cell transplant, relapse, and death.

Now the timeline form of these 3 subjects.

```
subset(mdata2, id<4)
```

```
##   id time trt sex flt3 sct cr relapse death
## 1  1    0  B  f   C  NA NA      NA    NA
## 2  1   44  B  f   C  NA  1      NA    NA
## 3  1  113  B  f   C  NA NA      1    NA
## 4  1  235  B  f   C  NA NA      NA     1
## 5  2    0  A  m   B  NA NA      NA    NA
## 6  2  200  A  m   B   1 NA      NA    NA
## 7  2  286  A  m   B  NA NA      NA     1
## 8  3    0  A  f   A  NA NA      NA    NA
## 9  3   38  A  f   A  NA  1      NA    NA
## 10 3 1983  A  f   A  NA NA      NA     0
```

In timeline data each row has a unique (subject identifier, time) pair, along with all the information that was observed at that time, with missing values for things that were not observed. Timeline data sets have many NA values. Static variables like treatment, sex and FLT3 state at randomization can be filled in for every row, as was done here, or not. (The analysis code can deal with either.) One could argue for instance that FLT3 was only observed at the start since a pathology assessment was not repeated, so should be NA on row 2, and certainly that none of these covariates were observed on the day of death.

For simple survival timeline style makes little sense: each subject will have to have two rows, one at 0 with the covariates and another at the death/censoring time with all covariates missing. For more complex data they have two advantages

- Counting process data sets often require the use of tmerge, which can be tricky. Timeline data can be built using more standard tools, which may be easier

- If one is rigorous with respect to the “observed at this time” rule for each line of data, most immortal time bias problems will be avoided

Example 1, NAFLD

The NAFLD data is found in 3 data sets. The original study contains all new NAFLD diagnosis in Olmsted County from 1997 to 2014, along with 4 potential controls for each case. Controls were selected matched on age, sex, and calendar year.

To protect patient confidentiality all time intervals are in days since the index date; none of the dates from the original data were retained. Subject age is their integer age at the index date, and the subject identifier is an arbitrary integer. As a final protection, the data includes only a 90% random sample of the original data.

- nafld1 has one row for each subject and contains the time fixed variables
- nafld2 contains time-dependent laboratory tests
- nafld3 contains the outcomes

```
nafld1[1:5,]
```

```
##   id age male weight height      bmi case.id futime status
## 1  1  57    0  60.0   163 22.69094   10630   6261      0
## 2  2  67    0  70.4   168 24.88403   14817    624      0
## 3  3  53    1 105.8   186 30.45354     3    1783      0
## 4  4  56    1 109.3   170 37.83010   6628   3143      0
## 5  5  68    1   NA    NA      NA    1871   1836      1
```

```
nafld2[1:5,]
```

```
##   id days test value
## 1  1 -459 hdl    75
## 2  1 -459 chol   75
## 3  1  183 hdl    64
## 4  1  183 chol   64
## 5  1 2030 hdl    74
```

```
nafld3[1:5,]
```

```
##   id days      event
## 1  3     0      nafl
## 2  4 -1287      htn
## 3  4 -1287 dyslipidemia
## 4  4 -1226    ang/isc
## 5  4  -90    ang/isc
```

```
table(nafld2$test)
```

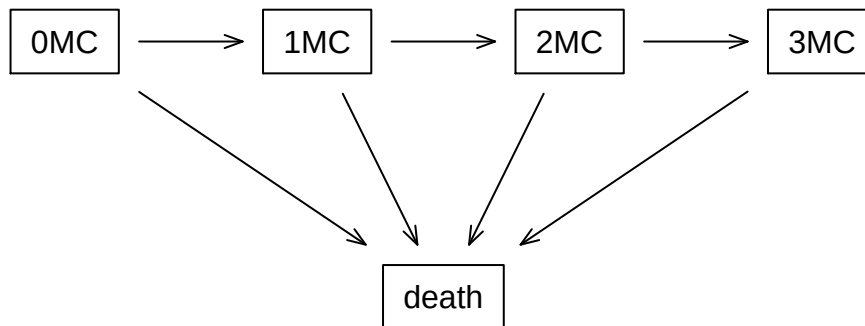
```
##
##   chol    dbp   fib4   hdl    sbp  smoke
## 161258 33430  3738 161259 33430  7008
```

```
table(nafld3$event)
```

```
##
##           afib           ang/isc cardiac arrest           diabetes heart failure
##           1935           2235           173           3452           1869
##           htn    dyslipidemia           MI           nafl           stroke
##           7097           10462           1199           3851           2054
```

The analysis in Allen 2018 looked at the accumulation of 3 metabolic comorbidities, leading to the 4 state model shown below. Age, male sex, NAFLD diagnosis, and BMI are the available baseline covariates. However, BMI is also a major risk factor for development of fatty liver. (We didn't include ongoing height/weight in the R data sets.)

```
nstate <- c("OMC", "1MC", "2MC", "3MC", "death")
nmat <- matrix(0, 5, 5, dimnames=list(nstate, nstate))
nmat[1,2] <- nmat[2,3] <- nmat[3,4] <- 1
nmat[1:4,5] <- 1
statefig(rbind(4,1), nmat)
```



In the nafld1 data set, the case.id variable, for a control, gives the id of the case they were matched with. For the incident cases, we set case.id = id. The nafld3 data also has nafld diagnosis, so check the consistency.

```
ndata <- nafld1 # working data set
ndata$nafld <- 1*(ndata$id == ndata$case.id) # create a variable
table(ndata$nafld)
```

```
##
##      0      1
## 14004  3514
```

```
ndx <- subset(nafld3, event=="nafld")
table(ndx$days==0)
```

```
##
## FALSE  TRUE
##   318   3533
```

```
subset(ndx, days>0)[1:5,]
```

```
##      id days event
## 20   13   83 nafld
## 86   53 1391 nafld
## 105  60 2272 nafld
## 233 137 2573 nafld
## 287 169   37 nafld
```

```
subset(ndata, case.id == 13)
```

```
##      id age male weight height      bmi case.id futime status nafld
## 3133  3138  55   0  99.7    173 33.31544      13   2498      0      0
## 6194  6200  54   0  62.0    155 25.65432      13   2425      0      0
## 7085  7092  54   0 118.6    170 41.15262      13   2432      0      0
## 15603 15619  54   0   NA     NA     NA      13   2353      0      0
```

```
temp <- subset(ndx, days==0 & !(id %in% ndata$id[ndata$nafld==1]))
dim(temp)
```

```
## [1] 19  3
```

```
temp[1:5,]
```

```
##      id days event
## 1575   805     0 nafld
## 6248  3158     0 nafld
## 6893  3472     0 nafld
## 7421  3742     0 nafld
## 10741 5452     0 nafld
```

```
subset(ndata, id %in% c(805, 3158))
```

```
##      id age male weight height      bmi case.id futime status nafld
## 804   805  58   1  81.8    175 26.62298    3859   5392      0      0
## 3153 3158  70   1  81.4    176 26.39826    2810   3492      1      0
```

```
subset(ndata, case.id == 805)
```

```
##      id age male weight height bmi case.id futime status nafld
## 3997   4002  57   1   NA     NA NA     805   4162      1      0
## 13359 13372  57   1   NA     NA NA     805   5233      0      0
## 14181 14194  57   1   NA     NA NA     805   1760      0      0
```

What is going on? 1. 318 subjects who were chosen as controls were found to have NAFLD later in life. You cannot exclude someone as a control at age 45 based on something that will happen to them some years later: this is a form of immortal time bias. When that control later became a case, 4 new controls for them were selected. 2. At this time NAFLD had a funny definition: if someone had fatty liver but also say a hepatitis C diagnosis, then they “didn’t have” NAFLD. The study question was more or less the consequences if NAFLD was the *only* liver disease. 3. What about hepatitis C 2 weeks later, 6 weeks later, ... The diagnostic process for someone with a GI complaint can sometimes stretch out for months, so the decision was that “real” NAFLD starts once 12 months has passed without another smoking gun. This is day 0. 4. Controls need to be matched on this new date, otherwise the NAFLD subjects will benefit from immortal time bias. 5. The 19 confuse me. Perhaps I made a mistake when going from dates to days? (It should be “days from initial recruitment” for all).

“If you don’t know where you are going, you might end up somewhere else”. Yogi Berra

What should we do? Different choices lead to a different data set and analysis changes. * Simple approach:

only follow the original enrollees and their controls. Ignore when someone changes the time-dependent covariate to NAFLD. * Treat NAFLD as a time-dependent covariate * Treat NAFLD as a time-dependent state * Restart: when a case becomes a control, split them into two copies: an original control and a new case.

Simple approach

For the simple analysis, we need a working data frame that has a row/days pair each time that the total number of MC increases. We can take advantage of the fact that each of the MC only is recorded at most once per subject: formally the variable is “history of diabetes”, etc. The lines below are likely an inefficient way to do this, but it works.

```
# first, subset to the relevant cases and controls
caseid <- subset(ndata, id==case.id)$id
tdata1 <- subset(ndata, case.id %in% caseid)
dim(tdata1) # this is the set subjects, one row each

## [1] 16253      10

# the endpoints of interest
enddata <- subset(nafld3, (id %in% tdata1$id) &
                  (event %in% c("diabetes", "dyslipidemia", "htn")))
enddata[1:10,]

##      id  days      event
## 2    4 -1287         htn
## 3    4 -1287 dyslipidemia
## 6    5   617         htn
## 7    7 -2685 dyslipidemia
## 8    9  -311      diabetes
## 10  10 -4498      diabetes
## 11  10 -4365 dyslipidemia
## 12  10 -3642         htn
## 13  11  -35         htn
## 14  12 -4723 dyslipidemia

table(enddata$days <=0)

##
## FALSE  TRUE
## 4644 14763

temp <- with(enddata, table(days <=0, event[,drop=TRUE]))
temp

##
##      diabetes  htn dyslipidemia
## FALSE      982 1964          1698
## TRUE      2219 4582          7962

round(temp/nrow(tdata1), 2)

##
##      diabetes  htn dyslipidemia
## FALSE      0.06 0.12          0.10
## TRUE      0.14 0.28          0.49

# counting process data
ndata2 <- tmerge(tdata1[c(1,2,3,6:7,10)], tdata1, id=id,
```

```

        death= event(futime, status),
        options=list(tstartname="day1", tstopname="day2"))
ndata2 <- tmerge(ndata2, enddata, id=id, mcount=cumtdc(days),
               mevent= cumevent(days))
subset(ndata2, id <8)

##   id age male      bmi case.id naflld day1 day2 death mcount mevent
## 1  1  57    0 22.69094   10630     0  0 6261     0     0     0
## 2  2  67    0 24.88403   14817     0  0  624     0     0     0
## 3  3  53    1 30.45354     3     1  0 1783     0     0     0
## 4  4  56    1 37.83010   6628     0  0 3143     0     2     0
## 5  5  68    1      NA   1871     0  0  617     0     0     1
## 6  5  68    1      NA   1871     0 617 1836     1     1     0
## 7  6  39    0 26.61559   7144     0  0 1581     0     0     0
## 8  7  49    0 25.51934   7507     0  0 3109     0     1     0

summary(ndata2)

## Call:
## tmerge(data1 = ndata2, data2 = enddata, id = id, mcount = cumtdc(days),
##        mevent = cumevent(days))
##
##           early late gap within boundary leading trailing tied missid
## death      0     0  0      0           0      0 16253     0     0
## mcount 14759     0  0  4637           0      4     7  535     0
## mevent 14759     0  0      0  4637           4     7  535     0

ndata2$istate <- factor(ndata2$mcount, 0:3, nstate[1:4])
temp <- ifelse(ndata2$death==1, 4, ndata2$mevent)
ndata2$event <- factor(temp, 0:4, c("censor", nstate[-1]))
check1 <- survcheck(Surv(day1, day2, event) ~1, data=ndata2,
                    id=id, istate=istate)
check1

## Call:
## survcheck(formula = Surv(day1, day2, event) ~ 1, data = ndata2,
##           id = id, istate = istate)
##
## Unique identifiers      Observations      Transitions
##           16253           20787           5825
##
## Transitions table:
##           to
## from      1MC  2MC  3MC death (censored)
## OMC      1738  68   4   244      5304
## 1MC        0 1733  27   233      4231
## 2MC        0   0  970  397      3375
## 3MC        0   0   0   411      2052
## death     0   0   0    0         0
##
## Number of subjects with 0, 1, ... transitions to each state:
##           count
## state      0     1     2     3     4
## 1MC      14515 1738     0     0     0
## 2MC      14452 1801     0     0     0

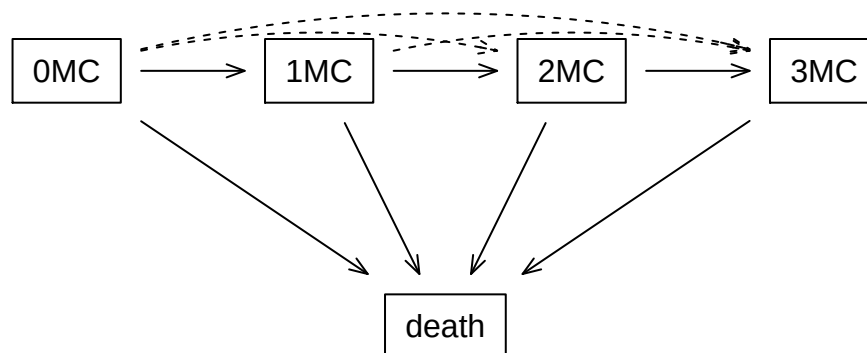
```

```
##    3MC    15252 1001    0    0    0
##    death 14968 1285    0    0    0
##    (any) 11726 3446 884 177 20
```

For an MSH model fit with `survfit` or `coxph` the underlying likelihood is event based. Each observed event contributes a term to the likelihood, and presumes * over the window of observation for a subject, all the transitions for that subject are recorded * the window of observation is complete, i.e., it includes all times at which an event, if it occurred, would have been recorded * non-informative censoring

One consequence of the first is that we do not need to inform the routine of which transitions can occur, and which are (effectively) absorbing: it tabulate them on its own. But, we need to check that it's assessment agrees with our own. In this case we have some jumps that were not in our original concept.

```
nmat <- matrix(0, 5, 5, dimnames=list(nstate, nstate))
nmat[1,2] <- nmat[2,3] <- nmat[3,4] <- 1
nmat[1:4,5] <- 1
nmat[1, 3:4] <- nmat[2,4] <- .7
statefig(rbind(4,1), nmat, alty=c(1,2,1,2,2,1,1,1,1,1))
```



To properly account for age we will do the primary analysis on age scale.

```
quantile(tdata1$age)
```

```
##    0%   25%   50%   75%  100%
##    18    42    52    62    98
```

```
ndata2$age1 <- ndata2$age + ndata2$day1/365.25
ndata2$age2 <- ndata2$age + ndata2$day2/365.25
```



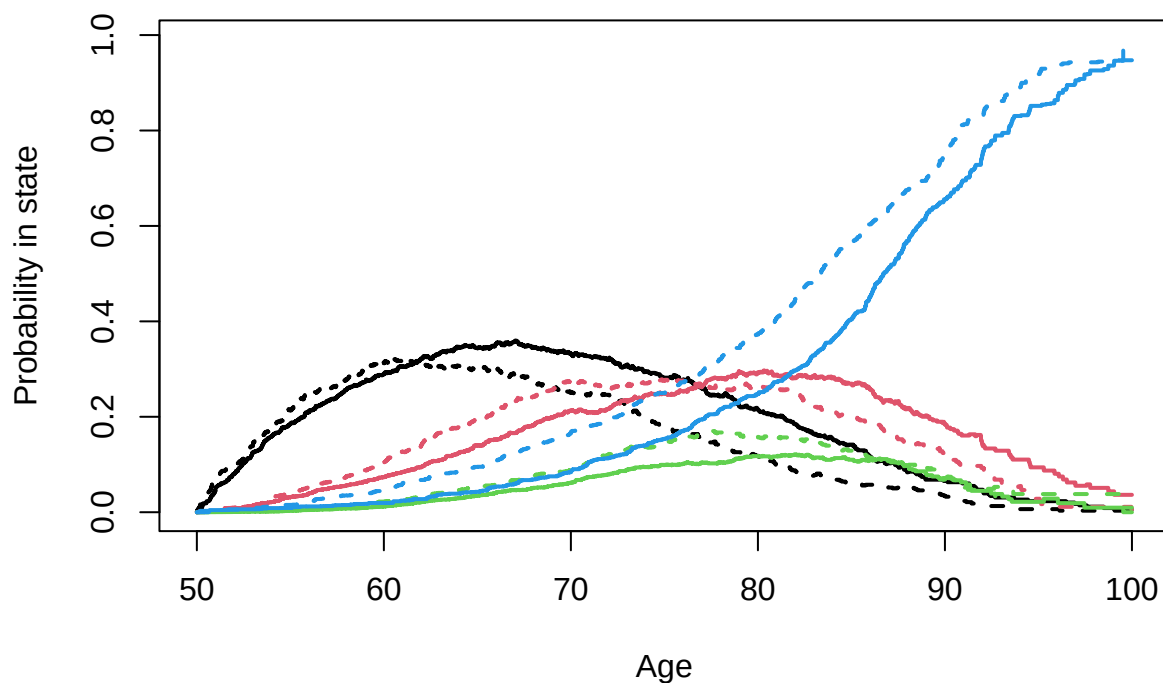
```
naj <- survfit(Surv(age1, age2, event) ~ male, ndata2, id=id, istate=istate,
               p0= c(1,0,0,0,0), start.time=50)
```

```
dim(naj)
```

```
## strata states
```

```
##      2      5
```

```
plot(naj[,2:5], lwd=2, lty=1:2, col=rep(1:4, each=2), xmax=100,
     xlab="Age", ylab="Probability in state")
```



```
#plot(naj[,1,], cumprob=2:5, col=1:4, yaxs='i', xmax=100,
#      xlab="Age", ylab="Probability in state")
#text(c(55, 68, 77, 79, 89), c(.8, .11, .37, .57, .69), nstate,
#      col=c(1,1,2,3,4))
```

```
dim(naj$cumhaz)
```

```
## [1] 13934    10
```

```
colSums(naj$n.transition)
```

```
## 1:2 1:3 2:3 1:4 2:4 3:4 1:5 2:5 3:5 4:5
## 1041 48 1317 4 19 780 218 215 386 402
```

The probability in state estimates are mostly oblivious to the presence of 0mc:2MC, 0MC:3MC and 1MC:3MC transtions, but the hazard estimates for those will be based on very small numbers. This affects the multi-state Cox model as well, since it is focused on hazards.

```

nacox1 <- coxph(Surv(age1, age2, event) ~ nafld + male,
                data=ndata2, id=id, istrate=istrate)

## Warning in agreg.fit(X, Y, istrat, offset, init, control, weights = weights, :
## Loglik converged before variable 8 ; beta may be infinite.

nacox1$cmap

##      1:2 1:3 2:3 1:4 2:4 3:4 1:5 2:5 3:5 4:5
## nafld  1   3   5   7   9  11  13  15  17  19
## male   2   4   6   8  10  12  14  16  18  20

nacox2 <- coxph(list(Surv(age1, age2, event) ~ nafld + male,
                    1:2 + 1:3 + 1:4 ~ nafld + male /common,
                    2:3 + 2:4 ~ nafld + male /common,
                    0:5 ~ male/common),
                data=ndata2, id=id, istrate=istrate)

nacox2

## Call:
## coxph(formula = list(Surv(age1, age2, event) ~ nafld + male,
##      1:2 + 1:3 + 1:4 ~ nafld + male/common, 2:3 + 2:4 ~ nafld +
##      male/common, 0:5 ~ male/common), data = ndata2, id = id,
##      istrate = istrate)
##
##
##      1:2, 1:3, 1:4      coef exp(coef) robust se      z      p
##      nafld 0.92706   2.52708   0.06932 13.374 < 2e-16
##      male  0.18656   1.20510   0.04940  3.776 0.000159
##
##
##      2:3, 2:4      coef exp(coef) robust se      z      p
##      nafld 0.52084   1.68344   0.05805  8.972 < 2e-16
##      male  0.21255   1.23683   0.05135  4.139 3.48e-05
##
##
##      3:4      coef exp(coef) robust se      z      p
##      nafld 0.48538   1.62479   0.07076  6.859 6.93e-12
##      male  0.11691   1.12401   0.06755  1.731  0.0835
##
##
##      1:5      coef exp(coef) robust se      z      p
##      nafld 0.68560   1.98497   0.23542  2.912  0.00359
##      male  0.33167   1.39329   0.05621  5.900 3.63e-09
##
##
##      2:5      coef exp(coef) robust se      z      p
##      nafld 0.53668   1.71033   0.15316  3.504 0.000458
##      male  0.33167   1.39329   0.05621  5.900 3.63e-09
##
##
##      3:5      coef exp(coef) robust se      z      p
##      nafld 0.51798   1.67863   0.10984  4.716 2.41e-06
##      male  0.33167   1.39329   0.05621  5.900 3.63e-09
##

```

```

##
## 4:5      coef exp(coef) robust se      z      p
##  nafld 0.03817  1.03891  0.10264 0.372    0.71
##  male  0.33167  1.39329  0.05621 5.900 3.63e-09
##
## States: 1= OMC, 2= 1MC, 3= 2MC, 4= 3MC, 5= death
##
## Likelihood ratio test=416.4 on 11 df, p=< 2.2e-16
## n= 20787, unique id= 16253, number of events= 5825
round(coef(nacox2, matrix=TRUE),3)

##      1:2  1:3  2:3  1:4  2:4  3:4  1:5  2:5  3:5  4:5
## nafld 0.927 0.927 0.521 0.927 0.521 0.485 0.686 0.537 0.518 0.038
## male  0.187 0.187 0.213 0.187 0.213 0.117 0.332 0.332 0.332 0.332
## attr("states")
## [1] "OMC" "1MC" "2MC" "3MC" "death"

nacox3 <- coxph(list(Surv(day1, day2, event)~ nafld+ bmi + strata(case.id),
                    1:2 + 1:3 + 1:4 ~ nafld + bmi /common,
                    2:3 + 2:4 ~ nafld + bmi / common),
                data=ndata2, id=id, istate=istate)
nacox3

## Call:
## coxph(formula = list(Surv(day1, day2, event) ~ nafld + bmi +
## strata(case.id), 1:2 + 1:3 + 1:4 ~ nafld + bmi/common, 2:3 +
## 2:4 ~ nafld + bmi/common), data = ndata2, id = id, istate = istate)
##
##
## 1:2, 1:3, 1:4      coef exp(coef) robust se      z      p
##  nafld 0.530875  1.700420  0.112369 4.724 2.31e-06
##  bmi   0.054123  1.055615  0.007847 6.897 5.30e-12
##
##
## 2:3, 2:4      coef exp(coef) robust se      z      p
##  nafld 0.346657  1.414332  0.078082 4.440 9.01e-06
##  bmi   0.039846  1.040650  0.006156 6.473 9.60e-11
##
##
## 3:4      coef exp(coef) robust se      z      p
##  nafld 0.416118  1.516065  0.094963 4.382 1.18e-05
##  bmi   0.042274  1.043180  0.009498 4.451 8.56e-06
##
##
## 1:5      coef exp(coef) robust se      z      p
##  nafld 0.28776  1.33344  0.38741  0.743 0.4576
##  bmi  -0.06494  0.93712  0.03504 -1.853 0.0638
##
##
## 2:5      coef exp(coef) robust se      z      p
##  nafld 0.29240  1.33964  0.23376 1.251 0.211
##  bmi   0.01357  1.01366  0.01841 0.737 0.461
##
##
## 3:5      coef exp(coef) robust se      z      p

```

```
##   nafld  0.34624   1.41374   0.14075  2.460 0.0139
##   bmi   -0.01249   0.98759   0.01120 -1.116 0.2646
##
##
## 4:5          coef exp(coef) robust se      z      p
##   nafld -0.003746  0.996261  0.146931 -0.025 0.980
##   bmi    0.009425  1.009469  0.012175  0.774 0.439
##
## States:  1= OMC, 2= 1MC, 3= 2MC, 4= 3MC, 5= death
##
## Likelihood ratio test=183.1 on 14 df, p=< 2.2e-16
## n= 15092, unique id= 11714, number of events= 4339
##   (5695 observations deleted due to missingness)
```

Because all the covariates are all time fixed, absolute risk estimates are fairly easy to get. For a Cox model, predictions are available for any a priori combination of covariates, which we provide in a dummy data set. Since it is multistate, we need to give a distribution of starting states, and since we are on age scale a starting time. (If you don't give these it will guess. I don't think I have every found the result of that to be useful.)

```
dummy <- expand.grid(male=0:1, nafld=0:1)
coxsurv2 <- survfit(nacox2, newdata=dummy, p0=c(1,0,0,0,0), start.time=45)
dim(coxsurv2)
```

```
##   data states
##      4      5
```

```
print(coxsurv2[1,], rmean=100)
```

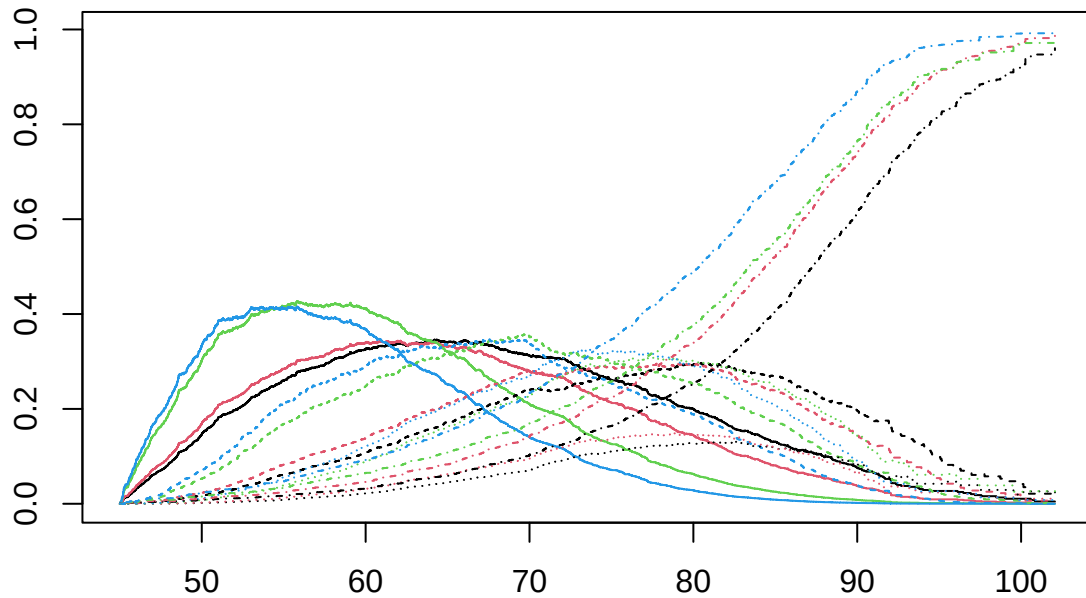
```
## Call: survfit(formula = nacox2, newdata = dummy, start.time = 45, p0 = c(1,
##   0, 0, 0, 0))
```

```
##
##          n nevent   rmean*
## OMC    16919      0 17.881230
## 1MC    16919    1291 10.500605
## 2MC    16919    1541  8.656491
## 3MC    16919     899  3.238753
## death 16919    1254 14.722922
##   *restricted mean time in state (max time = 100 )
```

```
xx <- paste0("male=", dummy$male, ", Nafld=", dummy$nafld)
sojourn <- summary(coxsurv2, rmean=100)$table
sojourn <- matrix(sojourn[, "rmean"], ncol=4, byrow=TRUE,
                  dimnames=list(nstate, xx))
rbind(sojourn, Total= colSums(sojourn))
```

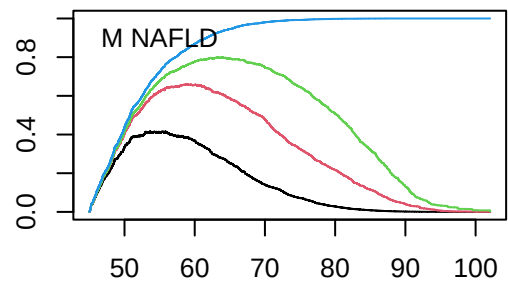
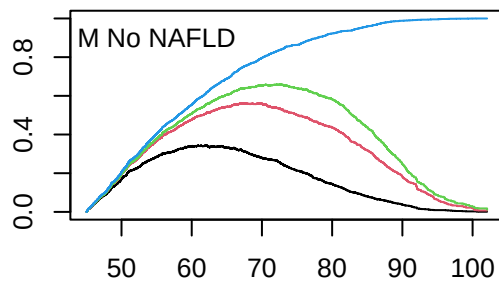
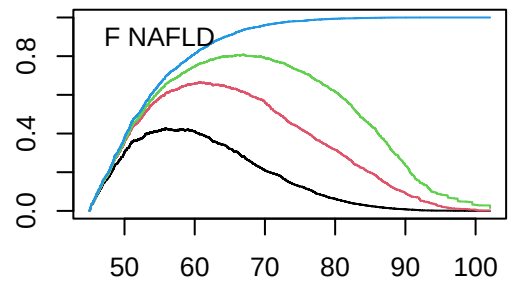
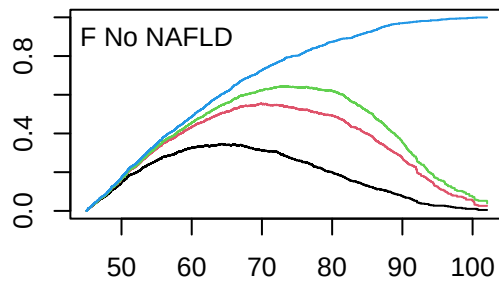
```
##      male=0, Nafld=0 male=1, Nafld=0 male=0, Nafld=1 male=1, Nafld=1
## OMC      17.881230      15.498161      8.885239      7.558078
## 1MC      10.500605      9.529256      9.558225      8.243057
## 2MC       8.656491      8.741058      9.801600      9.120698
## 3MC       3.238753      3.533408      7.997251      8.058475
## death    14.722922     17.698117     18.757685     22.019692
## Total    55.000000     55.000000     55.000000     55.000000
```

```
plot(coxsurv2[, -1], col=1:4, lty=rep(1:4, each=4))
```



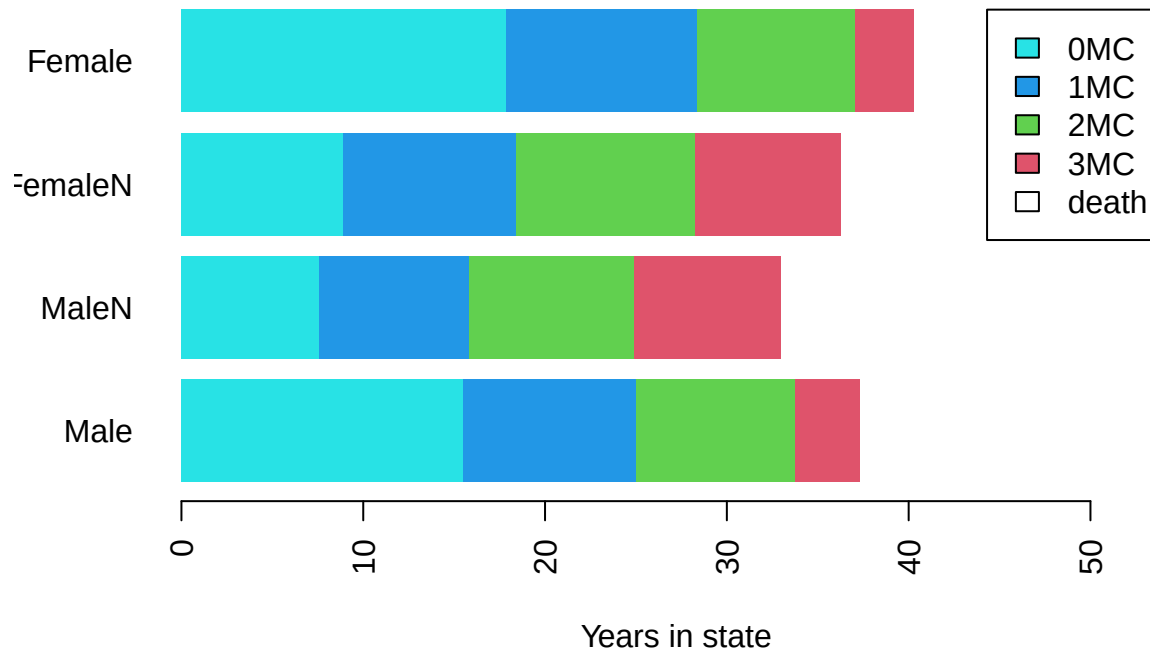
* The mean time in state has to add to $(100-45) = 55$, across the states. * Subscripting the curves is a useful tool * The simple plot has too many curves, it looks like a bowl of spaghetti. Better plotting requires artistic choices. * For a stratified model, the default plot will have a set of curves per stratum, which is not useful here.

```
opar <- par(mfcol=c(2,2), mar=c(5,5,2,1))
xx <- outer(c("F", "M"), c("No NAFLD", "NAFLD"), paste)
for (i in 1:4) {
  plot(coxsurv2[i,], cumprob=2:5, col=1:4, lty=rep(1:2, each=4))
  text(55, .9, xx[i])
}
```



```
par(opar)

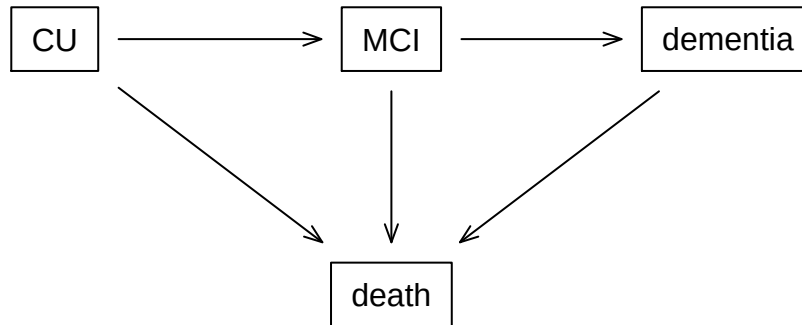
barplot(sojourn[,c(2,4,3,1)], horiz=TRUE, col=c(5:2, 0), las=2,
        names.arg=c("Male", "MaleN", "FemaleN", "Female"),
        xlab="Years in state", border=NA, legend=TRUE)
```



Mayo Clinic Study of Aging

The MCSA is based on a stratified random sample of subjects from Olmsted County, Minnesota. The dementia data set in the mshmm package is based on this data, and is able to reproduce Jack et al (2022) quite well, so we will use it for analysis. This data set is already in timeline form.

```
library(mshmm)
mstate <- c("CU", "MCI", "dementia", "death")
mmat <- matrix(0, 4, 4, dimnames=list(mstate, mstate))
mmat[1,2] <- mmat[2,3] <- 1
mmat[1:3,4] <- 1
statefig(rbind(3,1), mmat)
```



```
survcheck(Surv2(age, dx, repeated=TRUE) ~1, id=ptnum, dementia)$transitions
```

```
##           to
## from      CU  MCI dementia nondem off-dem death (censored)
## CU        13030 845      30   815    184    11      2234
## MCI        449 951     202   257    246    19      222
## dementia    0  29     163    0      0    172     30
## nondem      0  0      0      2      0    941     79
## off-dem     0  0      0      0      0    328    102
## death       0  0      0      0      0     0      0
```

```
survcheck(Surv2(age, dx) ~1, id=ptnum, data = dementia)
```

```
## Call:
## survcheck(formula = Surv2(age, dx) ~ 1, data = dementia, id = ptnum)
##
## Unique identifiers      Observations      Transitions
##           4317           21354           4528
##
## Transitions table:
##           to
## from      CU  MCI dementia nondem off-dem death (censored)
## CU         0 845      30   815    184    11      2347
## MCI        449  0     202   257    246    19      228
## dementia    0  29      0      0      0    172     30
## nondem      0  0      0      0      0    941     81
## off-dem     0  0      0      0      0    328    102
```



```
##      death      0      0      0      0      0      0      0
##
## Number of subjects with 0, 1, ... transitions to each state:
##          count
## state      0      1      2      3      4      5      6      7      9
##  CU      3912  363    40     2     0     0     0     0     0
##  MCI      3560  649   101     5     2     0     0     0     0
##  dementia 4094  214     9     0     0     0     0     0     0
##  nondem   3245 1072     0     0     0     0     0     0     0
##  off-dem   3887  430     0     0     0     0     0     0     0
##  death    2846 1471     0     0     0     0     0     0     0
##  (any)    2296  334 1125  380  123  47   9   2   1
```

We can see several things by looking at this table. * The survcheck routine was written for counting process data. One aspect of Surv2, which translates from timeline form to that form, is the repeated argument. When a subject has two sequential visits in the same state, the survfit and coxph routines need to know whether to treat the second observation as a new event. Most of the time they should not, repeated infections in the CGD data set is a counterexample, however. In this data each row is a visit, the repeated form clearly shows that over half of all visit pairs are CU:CU; most patients stay stable for many years before progression. The (any) row of the total visits table shows that 53% (2296/4327) have no recorded transtions during follow-up. * There are 30 instances of a CU:dementia pair. Per the model we assume that these subjects passed through MCI, we simply did assess them often enough to observe the transient MCI state. * Dementia:MCI is much less common than MCI:CU. This reflects on one hand the clinical reality that CU and MCI are more difficult to distinguish than dementia, and also fewer overall dementia cases. * There are no transitions from death to another state, or from death to censored (further follow-up after death). This is as it should be. * Off study dementia only transitions to death. This is because further record reviews are not scheduled for demented subjects. * There are no transitions from nondem to nondem or off-dem. This is, it turns out, an interesting artifact of how the online data collection system was designed. The nurse abstracters who review the history have a screen with fields for “last assessment date for dementia” and ‘dementia yes/no’.

At a second, third, etc review for a patient, should one occur, the contents of these are overwritten; thus the master data set at any given time only has the most recent value. * More dementias were found in record review than in study visits. We believe that this reflects informative censoring: subjects whose cognition is slipping selectively drop out, i.e., exactly those most likely to receive a dementia diagnosis at their next visit.

To reprise the 2022 paper we need to create a new endpoint of non-demented (ND) dementia, and death.

```
ddata <- dementia
table(ddata$dx)
```

```
##
##      alive      CU      MCI dementia      nondem      off-dem      death
##      2680    17264    2357      395      1074      430      1471
```

```
temp <- c(1, 2,2,3,2,3,4)[as.numeric(ddata$dx)]
ddata$dx2 <- factor(temp, 1:4, c("censor", "ND", "dementia", "death"))
with(ddata, table(dx2, dx))
```

```
##          dx
## dx2      alive      CU      MCI dementia      nondem      off-dem      death
##  censor    2680      0      0      0      0      0      0
##    ND        0 17264  2357      0    1074      0      0
##  dementia    0      0      0     395      0     430      0
##    death    0      0      0      0      0      0    1471
```

```
survcheck(Surv2(age, dx2) ~1, ddata, id=ptnum)
```

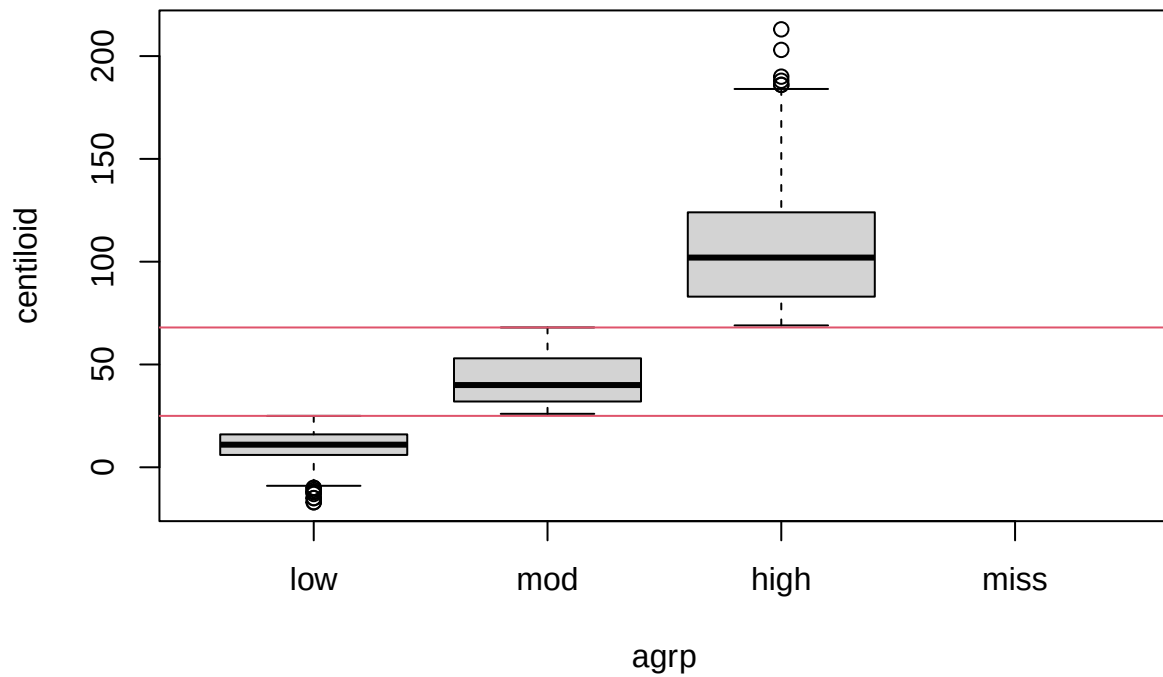
```
## Call:
```

```
## survcheck(formula = Surv2(age, dx2) ~ 1, data = ddata, id = ptnum)
##
## Unique identifiers      Observations      Transitions
##           4317           21354           2162
##
## Transitions table:
##           to
## from      ND dementia death (censored)
## ND         0       662   971       2713
## dementia   29        0   500       132
## death      0        0    0        0
##
## Number of subjects with 0, 1, ... transitions to each state:
##           count
## state      0    1    2    3    4
## ND        4288   29    0    0    0
## dementia  3664   644    9    0    0
## death     2846  1471    0    0    0
## (any)     2707  1088  499   16    7
```

The 2022 analysis was actually for time to *first* dementia. This change to using only first event is common enough that we have added an option to the survival package. `survcheck(Surv2(age, dx2, repeated='first') ~1, ddata, id=ptnum)`

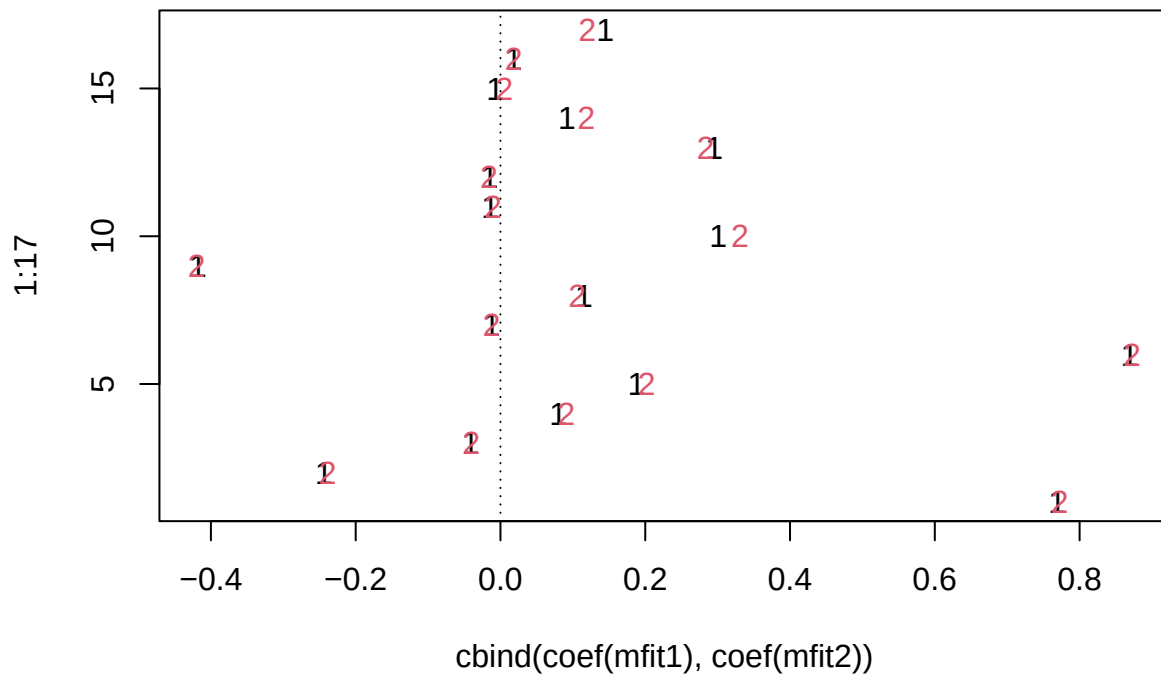
```
ddata$edu4 <- pmax(8, ddata$educ) / 4
check2 <- survcheck(Surv2(age, dx2, repeated='first') ~1, ddata, id=ptnum)
check2$transition
```

```
##           to
## from      dementia death (censored)
## ND         653   957       2707
## dementia    0   514       138
## death       0    0        0
# add categorized amyloid, and initial CMC,
# but we want lvcf to work correctly wrt amyloid
atemp <- with(ddata, lvcf(ptnum, centiloid))
temp <- ifelse(is.na(atemp), 4,
               as.numeric(cut(ddata$centiloid, c(-50, 25, 68, 500))))
ddata$agrp <- factor(temp, 1:4, c("low", "mod", "high", "miss"))
findex <- match(ddata$ptnum, ddata$ptnum)
ddata$icmc <- ddata$cmc[findex]
boxplot(centiloid ~ agrp, ddata)
abline(h=c(25, 68), col=2)
```



Now fit two models, using CMC and initial CMC.

```
mfit1 <- coxph(list(Surv2(age, dx2) ~ male + any.e4 + educ + cmc,
  1:2 ~ male*any.e4),
  data= ddata, id=ptnum)
mfit2 <- coxph(list(Surv2(age, dx2) ~ male + any.e4 + educ + icmc,
  1:2 ~ male*any.e4),
  data= ddata, id=ptnum)
matplot(cbind(coef(mfit1), coef(mfit2)), 1:17)
abline(v=0, lty=3)
```



How do I get a probability in state curve for a 65 year old female, e4-, cmc of 2, current state of ND, and *current* centiloid of 30? The standard survfit call doesn't work for TD covariates. Answer, make more states.

```
with(ddata, table(agrp, dx2))
```

```
##      dx2
## agrp  censor   ND dementia death
## low    2  1994    11      0
## mod    1   484     4      0
## high   0   433    26      0
## miss  1473 13315   594  1166
```

```
temp1 <- with(ddata, lvcf(ptnum, ifelse(dx2=="censor", NA, dx2)))
temp1 <- factor(temp1, 2:4, c("ND", "dementia", "death"))
with(ddata, table2(agrp, temp1))
```

```
##      temp1
## agrp      ND dementia death
## low   1996     11      0
## mod   485      4      0
## high  433     26      0
## miss 14712    670  1166
## <NA>  5617    246   305
```

```
temp2 <- with(ddata, as.numeric(agrp) + 4* as.numeric(temp1) -4)
dx3 <- factor(temp2, c(0:8, 12),
              c("censor",
                outer(c("low", "mod", "high", "miss"), c("ND", "dementia"),
```

```

      paste, sep='_'), "death"))
check3 <- survcheck(Surv2(age, dx3) ~ 1, id=ptnum, ddata)
check3$trans

```

```

##           to
## from      low_ND mod_ND high_ND miss_ND low_dementia mod_dementia
## low_ND      0   108     2     0         7         1
## mod_ND      9     0    65     0         0         0
## high_ND     0     1     0     0         0         0
## miss_ND    570   130   144     0         1         3
## low_dementia 1     0     0     0         0         0
## mod_dementia 0     0     0     0         0         0
## high_dementia 0     0     2     0         0         0
## miss_dementia 0     1     0    17         2         0
## death       0     0     0     0         0         0

```

```

##           to
## from      high_dementia miss_dementia death (censored)
## low_ND              0             0     0      1054
## mod_ND              0             0     0      253
## high_ND             16             0     0      248
## miss_ND              5           480    782     1480
## low_dementia         0             0     0         9
## mod_dementia         0             0     0         4
## high_dementia        0             0     0        19
## miss_dementia        0             0    384        76
## death               0             0     0         0

```

```
dim(check3$trans)
```

```
## [1]  9 10
```

```
table(check3$trans[, -10] > 0)
```

```

##
## FALSE  TRUE
##    59    22

```

```
table(check3$trans[, -10])
```

```

##
##  0  1  2  3  5  7  9 16 17 65 108 130 144 384 480 570 782
## 59  5  3  1  1  1  1  1  1  1  1  1  1  1  1  1  1

```

This is as far as I got.